# Statistical analysis plan (SAP)

## Write the analysis down before you see the data

Foundations of Clinical Research · Sessions 5–6 (6.4) · Template version 1.0

### How to use this template

- Replace every [grey italic text in brackets] with your own text. Press Ctrl + H and search for "[" to step through them, and check none remain before you share the document.
- Grey italic guidance and shaded "Example" boxes are there to help you. Delete them as you write: Home → Styles → right-click "Guidance" (or "Example") → Select All → Delete.
- Finish and sign the SAP before the database is locked (and, in a trial, before unblinding). Analyses that were not planned can still be done, but must be reported as exploratory.
- The examples use the running example as a two-arm trial: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?* Notes marked "Observational study" show what changes for a cohort.
- Write it with your statistician. In this course you need to understand each choice; the instructor runs the analyses.
- **Worked example:** a complete SAP for the course dataset (FLUID-ICU, positive fluid balance by day 3 and 28-day mortality) is in Course/Data/SAP_FLUID-ICU.md. Compare your plan with it.

| | |
|---|---|
| **Study title:** | *[Full title]* |
| **Protocol version:** | *[Version and date]* |
| **SAP version / date:** | *[1.0] · [DD/MM/YYYY]* |
| **Written by:** | *[Name(s) and role(s)]* |

## 1 Study summary and objectives

*Copy from the protocol: design, population, intervention/exposure and comparator, and the primary and secondary objectives. Every analysis below must link to one objective.*

**Example:** *Pilot randomised trial, 1:1, adults with septic shock. Primary objective: compare 28-day all-cause mortality between a restrictive fluid strategy and usual care. Secondary: cumulative fluid balance at day 5; renal replacement therapy; time to death up to day 28.*

*[Summary and objectives]*

# 2 Outcomes and their definitions

*Define each outcome exactly as it will be computed from the database: variable name, type, unit and time point. One primary outcome.*

| Outcome | Type | Definition (variables) | Time point |
|---|---|---|---|
| *Primary: all-cause mortality* | *Binary* | *vital_d28 = 1 (dead)* | *Day 28* |
| *Cumulative fluid balance* | *Continuous (ml/kg)* | *cfb_kg_d5 = Σ(fin − fout) days 1–5 ÷ weight* | *Day 5* |
| *Time to death* | *Time-to-event* | *death_dt − enrol_dt; censored at day 28* | *Up to day 28* |
| *[Outcome]* | *[Type]* | *[Definition]* | *[Time point]* |
| *[Outcome]* | *[Type]* | *[Definition]* | *[Time point]* |

# 3 Analysis populations

*Who is included in each analysis. Trials: the intention-to-treat (ITT) population is everyone randomised, analysed in the group they were allocated to. Observational studies: everyone meeting the eligibility criteria, with reasons for any exclusion in the flow diagram.*

| Population | Definition | Used for |
|---|---|---|
| **Intention-to-treat (ITT)** | *[All randomised participants, as allocated]* | *[Primary analysis]* |
| **Per-protocol** | *[Participants who received the allocated strategy as defined in the protocol]* | *[Sensitivity analysis]* |
| **Safety** | *[All participants who received any study intervention, as treated]* | *[Adverse events]* |

# 4 Descriptive analysis (Table 1)

*Describe baseline characteristics by group (see the Table 1 shell in section 13). Continuous variables: mean (SD) if roughly symmetric, otherwise median (IQR). Categorical: n (%). In a trial, do NOT test baseline differences with p-values: any difference is due to chance. Report the number with missing data for each variable.*

> **Example:** *Age, sex, weight, source of infection, comorbidities, SOFA score and lactate summarised per arm; lactate is skewed, so median (IQR).*

*[Our descriptive plan]*

# 5 Primary analysis

*One analysis, fully specified: the test or model, the effect measure with its 95% confidence interval, any adjustment, and the significance level. The p-value supports the effect estimate; it does not replace it.*

**Outcome:** *[Primary outcome and population]*

**Method:** *[Test or model]*

**Effect measure:** *[e.g. risk ratio, risk difference, odds ratio, hazard ratio, mean difference]*

**Adjustment:** *[Variables, and why]*

**Significance:** *[Two-sided, 5% level]*

> **Example:** *28-day mortality in the ITT population compared with a chi-square test. Effect reported as risk ratio and risk difference with 95% CI. Supportive analysis: logistic regression adjusted for the stratification variable (baseline lactate > 4 mmol/L), reported as odds ratio (95% CI).*

# 6 Secondary analyses

*For each secondary outcome: summary, test or model and effect measure. Say clearly that secondary results are supportive or exploratory.*

> **Example:** *Fluid balance: median (IQR) per arm, Mann–Whitney U test, difference in medians with 95% CI. Time to death: Kaplan–Meier curves, log-rank test, Cox model hazard ratio (95% CI), proportional hazards checked. Renal replacement therapy: risk ratio (95% CI).*

*[Our secondary analyses]*

# 7 Confounders and adjustment

*Trials: adjust only for stratification variables and a few pre-specified strong prognostic factors. Observational study: list the confounders chosen IN ADVANCE from subject knowledge (a simple causal diagram helps), not by stepwise selection. Keep about 10 outcome events per variable in the model.*

> **Example:** *Observational version of the running example: sicker patients receive more fluid AND die more (confounding by indication), so the model adjusts for age, SOFA score, lactate and source of infection. With about 68 expected deaths, the model can take at most 6–7 variables.*

*[Our adjustment plan]*

# 8 Subgroups and interaction (pre-specified)

*List only a few subgroups, with the expected direction and a reason. Test whether the effect differs with an interaction test, not by comparing p-values in each subgroup. Present a forest plot and call the results exploratory.*

> **Example:** *Baseline lactate (≤ 4 vs > 4 mmol/L) and SOFA score (≤ 10 vs > 10): treatment × subgroup interaction term in the logistic model.*

*[Our subgroups]*

# 9 Multiple comparisons

*Many tests mean some "significant" results by chance (the jelly-bean problem). Protect yourself with ONE primary outcome. Say how secondary outcomes are handled: interpreted as exploratory without adjustment, or adjusted (e.g. Holm or Bonferroni).*

> **Example:** *One primary outcome, tested at 5%. Secondary outcomes reported with 95% CIs and interpreted as exploratory; no adjustment for multiplicity.*

*[Our approach]*

## 10 Missing data

*Plan four things. (1) How missingness will be assessed and reported: n (%) missing per variable and by group. (2) The likely mechanism: missing completely at random (MCAR), at random given observed data (MAR), or not at random (MNAR). (3) The main approach: complete-case analysis, or multiple imputation if more than about 5% is missing and MAR is plausible. (4) A sensitivity analysis that tests the assumption.*

**Assessment:** *[How missingness is described and explored]*

**Main approach:** *[Complete-case / multiple imputation (variables in the model, number of imputations)]*

**Sensitivity:** *[e.g. best-case / worst-case, or a different imputation model]*

> **Example:** *Vital status at day 28 is expected to be missing for < 2% (phone follow-up). Main analysis: complete case. Sensitivity: missing patients assumed dead (worst case) and alive (best case). Baseline lactate, if > 5% missing, multiply imputed (20 imputations) from age, SOFA and arm.*

## 11 Sample size

*Copy the calculation from the protocol and state every assumption with its source: expected value in the comparison group, smallest important difference, power, significance level, allowance for loss.*

> **Example:** *40% vs 30% mortality, 80% power, two-sided 5%: 356 per group, about 790 in total with 10% loss. The pilot (60 patients) is not powered for mortality; it estimates recruitment and adherence.*

*[Our sample size]*

## 12 Software

*Name the software and version, and where the analysis code will be kept.*

*[e.g. R version 4.x (packages ...) / Stata / SPSS; code stored in [location] with version control]*

# 13 Table and figure shells

*Draw the empty tables and figures now. They show exactly what will be calculated, and they become the tables of the paper. Use n (%) and median (IQR) with units, n per column and footnotes.*

### Table 1. Baseline characteristics by group

| Characteristic | Restrictive (n = __) | Usual care (n = __) |
|---|---|---|
| Age, years, mean (SD) | | |
| Female sex, n (%) | | |
| SOFA score, median (IQR) | | |
| Lactate, mmol/L, median (IQR) | | |
| Source of infection: lung, n (%) | | |
| *[Characteristic, unit, summary]* | | |

Footnote: SD, standard deviation; IQR, interquartile range. Missing data: [n] for lactate.

### Table 2. Primary and secondary outcomes

| Outcome | Restrictive | Usual care | Effect (95% CI) | p |
|---|---|---|---|---|
| 28-day mortality, n (%) | | | RR __ (__ to __) | |
| Cumulative fluid balance day 5, ml/kg, median (IQR) | | | Difference __ (__ to __) | |
| Renal replacement therapy, n (%) | | | RR __ (__ to __) | |

### Figures

- Figure 1. Participant flow diagram (CONSORT for trials, STROBE for observational studies).
- Figure 2. Kaplan–Meier curves of survival to day 28 by group, with numbers at risk.
- Figure 3. Forest plot of pre-specified subgroups (if any).

# 14 Version history and sign-off before data lock

*Sign before the database is locked (and before unblinding in a trial). Any later change gets a new version number, the date and the reason, and is reported in the paper.*

| Version | Date | Changes and reason |
|---|---|---|
| *[1.0]* | *[DD/MM/YYYY]* | First version |
| *[1.1]* | *[DD/MM/YYYY]* | *[Change and reason]* |

| Role | Name | Signature | Date (DD/MM/YYYY) |
|---|---|---|---|
| Statistician | | | |
| Principal investigator | | | |
| Other (e.g. trial steering committee) | | | |

Database lock date: *[DD/MM/YYYY]*